

Introduction:

Hi, my name is Tharun Kumar Vaspari, and I'm currently pursuing my Master's in Computer Science at Clemson University. I'm passionate about building secure and scalable systems and enjoy working on both development and cloud-based infrastructure.

I'm currently working as a Graduate Teaching Assistant for the Cloud Computing Architecture course, where I mentor students in designing cloud solutions using AWS. I guide them on setting up virtual environments, securing deployments, and implementing monitoring tools like CloudWatch and CloudTrail. This experience has strengthened my understanding of real-world system architecture and platform management.

Before this, I worked as a Full-Stack Developer Intern at Attic Infomatics, where I developed a flight booking platform. I worked on both the frontend using React and Angular, and backend using Spring Boot, MySQL, and MongoDB. I also optimized API performance and improved the user experience across devices.

In my academic journey, I've completed several hands-on projects, such as Enhancing Data Security in the Cloud with Semi-Trusted Third-Party Involvement using AES encryption and SMTP for secure data sharing, a currency recognition system using CNN in Python, and a secure watermarking mechanism using cryptographic techniques. These projects reflect my interest in combining development with data security and AI.

I've also completed certifications like Microsoft Azure Fundamentals (AZ-900) and courses in PostgreSQL, Java, and Google Cloud, which helped me strengthen both my theoretical and practical skills.

I'm now looking for internship opportunities where I can apply my skills in cloud platforms, and software development to contribute to impactful, real-world applications.

TA Project:

As a Graduate Teaching Assistant for the Cloud Computing Architecture course at Clemson University, I am assisting on a project called TigerMed - a cloud-based healthcare application, which simulates a healthcare platform using AWS. The goal was to design a secure and scalable cloud system to manage patient data and doctor appointments.

We used Amazon EC2 for hosting the web and application servers, with Auto Scaling and Load Balancers to handle traffic during busy times. Amazon S3 was used to store uploaded documents securely, and AWS Lambda helped automate tasks like extracting text from documents or suggesting appointment types.

To keep everything secure, we used IAM roles and Security Groups to control access between the different parts of the system. Each environment—Development, Testing, and Production—was set up in its own VPC to keep data separate and safe.

We also added monitoring tools like CloudWatch and CloudTrail to track system health and log user activity. I helped guide students through designing this setup, solving issues, and making sure the project followed real-world cloud security and performance best practices.

Internship:

During my internship at **Attic Infomatics**, I worked as a **Full-Stack Developer Intern** on an aviation flight booking platform. I started with the **frontend**, using **HTML**, **CSS**, **JavaScript**, and later integrated **React**, **Angular**, and **Bootstrap** to build a responsive interface — which led to a **25% boost in user engagement**.

On the **backend**, I developed **RESTful APIs** using **Spring Boot** and **PHP**, connected to **MySQL** and **MongoDB** for managing structured and flexible airline data. I optimized API responses and database queries, reducing load times by **30%**.

We normalized incoming flight data from three different airlines using custom scripts and added **form validation** and **session handling** for better security and user experience. Working in an **Agile environment** with **GitHub** for collaboration, we delivered a functional booking system that supported **500+ users** and was ready for deployment.

This project gave me in-depth experience in **frontend and backend integration**, **API development**, **data normalization**, and working within a **collaborative, production-level software development process**.

Cloud Project:

The project titled "Enhancing Data Security in the Cloud with Semi-Trusted Third-Party Involvement" focuses on securing cloud-stored data in environments where the service provider or auditor is not fully trusted. With the increasing reliance on cloud platforms, it's essential to ensure that both data integrity and user privacy are maintained, especially when data is shared or managed by external systems. This project was built to provide a privacy-preserving public auditing mechanism that enables users to verify the security of their data without revealing its content to third-party auditors.

The system was designed around four main modules: encryption, integrity checking, key transfer, and data retrieval. In the encryption module, I used the Advanced Encryption Standard (AES) with a 128-bit key to encrypt all files before uploading them to the cloud. This ensures that even if someone accesses the stored data, they cannot understand or misuse it without the decryption key.

To verify that data hasn't been tampered or corrupted, the integrity checking module was developed using the MD5 hashing algorithm. When a file is uploaded, its hash is generated and stored. During retrieval, the system generates the hash again and compares it with the original to ensure data integrity. This step confirms that the file has not been altered during storage or transmission.

The key transfer module securely shares encryption keys using the SMTP protocol through email. Instead of exposing keys directly, they are sent in a controlled and encrypted format, ensuring that only the intended user can decrypt and access the original data. Finally, the data retrieval module enables users to access and decrypt their files once their identity is verified. This includes rechecking the integrity using MD5 and then decrypting the file using the secure key.

Overall, this project successfully secured over 500 GB of simulated cloud data, achieving strong encryption and data validation. It demonstrated an improvement in data security, based on simulated attack scenarios. The solution is modular, scalable, and applicable to domains like healthcare, banking, or academic cloud systems where data privacy is critical.

Azure Certificate:

The Microsoft Certified: Azure Fundamentals (AZ-900) certification provided me with a solid understanding of core cloud concepts and Microsoft Azure services. Through this course, I learned the foundational principles of cloud computing, including the differences between IaaS, PaaS, and SaaS, and how to choose the right model for different business scenarios.

I gained hands-on knowledge of key Azure services such as Azure Compute, Storage, Networking, and Databases, and how they are used to build and manage scalable cloud applications. I also learned about Azure Resource Manager, virtual machines, blob storage, and Azure SQL databases.

In addition to services, the certification focused on Azure's security, privacy, compliance, and trust models. I explored tools like Azure Security Center and Azure Defender, and understood how role-based access control (RBAC) and network security groups (NSGs) help in securing cloud environments.

The course also covered pricing and support plans, giving me the ability to estimate costs using the Azure Pricing Calculator and understand how to optimize cloud spending based on usage.

Overall, this certification helped me build a strong foundation in cloud computing, which I've been able to apply to real-world projects involving AWS and Azure, particularly in managing infrastructure, implementing access controls, and optimizing deployment environments.

1. Why do you think you are suitable for this role?

I believe I'm a strong fit for this role because, as a **Teaching Assistant**, I've designed and implemented **cloud-based systems using AWS**, focusing on security, monitoring, and automation. I've worked hands-on with services like **EC2, S3, Lambda, IAM, VPC, CloudWatch, and CloudTrail** to build scalable and secure environments. I also guided students in configuring **multi-tier architectures**, managing **access control policies**, and setting up **automated monitoring** — all of which align closely with the responsibilities of managing platforms and supporting multiple environments at AgFirst.